

S. Earnest Kirubakaran Oswarld

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Summary

AI and Data Science student specializing in Generative AI, NLP, and applied machine learning, with hands-on experience building LLM-powered applications, FastAPI backends, and data dashboards.

Education

Karunya Institute of Technology and Sciences

B.Tech, Artificial Intelligence & Data Science

Tamil Nadu, India

Expected 2027

AVB Matriculation Higher Secondary School

Higher Secondary Education (HSE)

Tamil Nadu, India

2023

Skills

Programming: Python, SQL, PostgreSQL, React, HTML, CSS, JavaScript

AI/ML: Machine Learning, LLMs, Generative AI, NLP

Data Analysis: Pandas, NumPy, EDA, Data Visualization

Tools: Power BI, Tableau, Git, GitHub, VS Code

IoT: Raspberry Pi, Arduino, Sensor Interfacing

Experience

Edify Techno Solutions

Data Science & Data Analytics Intern

Chennai, India

June 2025 – July 2025

- Built interactive Power BI dashboards analyzing a 9,995-record retail dataset across 4 regions and 3 product categories.
- Developed KPI metrics and drill-down visualizations used by HR and business teams for sales and profit performance tracking.
- Performed data cleaning, transformation, and modeling using Power Query and DAX, standardizing reporting calculations.
- Automated monthly reporting dashboards, reducing manual Excel consolidation effort and improving reporting efficiency.

Projects

AI-Based Architecture Generator | *Python, FastAPI, React Flow, PyTorch, LoRA*

github.com/earnest-s/archgen

- Built a local-first AI application that converts natural language prompts into editable software architecture diagrams.
- Developed a FastAPI backend with Transformer-based inference and structured JSON validation for architecture generation.
- Engineered a React Flow editor with drag-and-drop editing, local persistence, theme support, and PNG export.
- Fine-tuned Qwen2.5-1.5B-Instruct using QLoRA and optimized inference with PyTorch, PEFT, and BitsAndBytes 4-bit quantization, enabling the model to run entirely on a consumer laptop GPU without cloud dependency.

AIRA – Multi-Agent AI Decision System | *Hackathon Project, Python*

github.com/SparkWorks-Spark-your-Ideas/AIRA

- Designed backend architecture for a multi-agent AI framework coordinating specialized agents across finance, risk, and compliance decision workflows.
- Engineered agent communication pipelines and domain-specific analysis modules generating consolidated, explainable recommendations from structured inputs.
- Built scalable REST APIs and data-processing services enabling real-time reasoning traceability and dashboard-ready outputs.

JSON Prompt Generator | *FastAPI, Mistral AI, React, Collaboration*

github.com/MSG.../JSON-Prompt-Generator

- Built a full-stack AI application converting free-text input into structured, context-aware JSON prompts.
- Integrated Pydantic AI with Mistral AI on the backend for schema validation and automatic error recovery.
- Built a React and Tailwind frontend with instant conversion, JSON copy-to-clipboard, and history tracking.

Certifications

Microsoft Certified: Azure AI Fundamentals – Microsoft, 2026

Developing Machine Learning Solutions – Amazon Web Services (AWS), 2026

Introduction to Model Context Protocol – Anthropic, 2026

Getting Started with AI on Jetson Nano – NVIDIA, 2025